WORKED SOLUTION

STEP III 1993, Question 6

Pure • Loci in the Argand Diagram

STEP III 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q6

The key idea

For a circle, $|z-k|=r$ is the natural description, and squaring it with $|u|^2=uu^*$ turns it into an equation in z and z^* with no square roots. Once a locus is written that way, a Möbius map such as $w=z/(z-1)$ is easy to handle: invert it to get z in terms of w, substitute, clear denominators, and read off the new centre and radius by comparing with the general form.

Solution

The equation of a circle

P lies on the circle exactly when its distance from K is r, that is $|z-k|=r$. Squaring, and using $|u|^2=uu^*$ with $(z-k)^*=z^*-k^*$,

$$r^2=(z-k)(z^*-k^*)=zz^*-kz^*-k^*z+kk^*,$$

so $zz^*-kz^*-k^*z+kk^*-r^2=0$, as required.

Conversely, any equation $ww^*-cw^*-c^*w+cc^*-s^2=0$ with $s>0$ says $|w-c|^2=s^2$, so it is the circle with centre c and radius s. We use this to recognise circles below.

The locus of P

The circle on OA as diameter has centre i (the midpoint of 0 and $2i$) and radius 1. With $k=i$, $k^*=-i$, $kk^*=1$ and $r=1$, the first part gives

$$zz^*+iz-iz^*=0. \qquad (*)$$

The locus L of Q_1

Note that $z = 1$ is not on the circle (its distance from i is $\sqrt{2}$), so $w_1 = z/(z-1)$ is defined for every P. Write w for w_1. Solving $w(z-1) = z$ for z gives

$$z = \frac{w}{w-1}, \qquad z^* = \frac{w^*}{w^*-1},$$

and $w \neq 1$, because $w = 1$ would need $z = z - 1$. Substituting into $(*)$:

$$\frac{ww^*}{(w-1)(w^*-1)} + \frac{iw}{w-1} - \frac{iw^*}{w^*-1} = 0.$$

Multiplying through by $(w-1)(w^*-1)$, which is non-zero,

$$\begin{aligned} 0 &= ww^* + iw(w^*-1) - iw^*(w-1) \\ &= ww^* + iww^* - iw - iww^* + iw^* \\ &= ww^* - iw + iw^*. \end{aligned}$$

So L has equation

$$w_1 w_1^* - iw_1 + iw_1^* = 0.$$

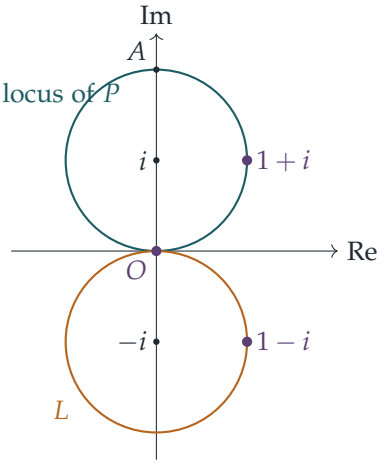
To identify it, compare with the general form: we need $-k^* = -i$ and $-k = i$, so $k = -i$; then $kk^* = 1$, and the constant term $kk^* - r^2$ is 0, so $r = 1$. Hence L is the circle with centre $-i$ and radius 1, i.e. $|w_1 + i| = 1$. It is the reflection of P's circle in the real axis, and it also passes through the origin.

Ask yourself Is every point of this circle actually reached, or only part of it?

Every step above can be reversed. If $w \neq 1$ lies on $|w + i| = 1$, then $z = w/(w-1)$ satisfies $(*)$, so it is a point P on the circle with $w_1 = w$. The only point that could be missed is $w = 1$, and that is not on L, since $|1 + i| = \sqrt{2}$. So the locus is the whole circle.

The locus of Q_2

Taking the conjugate of $|z - i| = 1$ gives $|z^* + i| = 1$, since $|u^*| = |u|$ and $(z-i)^* = z^* + i$. So $w_2 = z^*$ satisfies $|w_2 + i| = 1$. As z runs over the whole circle centred at i, z^* runs over the whole circle centred at $-i$. Equivalently, conjugating $(*)$ gives $z^*z - iz^* + iz = 0$, which is the equation of L with $w = z^*$. So the locus of Q_2 is also L.



When Q_1 and Q_2 coincide

We need $\dfrac{z}{z-1} = z^*$, that is

$$z = zz^* - z^*.$$

On the circle, $(*)$ gives $zz^* = iz^* - iz$. Substituting,

$$z = iz^* - iz - z^* \quad \Longrightarrow \quad (1+i)z = (i-1)z^*.$$

One solution is $z = 0$. Otherwise $z^* \neq 0$ and

$$\frac{z}{z^*} = \frac{i-1}{1+i} = \frac{(i-1)(1-i)}{(1+i)(1-i)} = \frac{2i}{2} = i.$$

Writing $z = \rho e^{i\theta}$ with $\rho > 0$, this says $e^{2i\theta} = i$, so $\theta = \frac{\pi}{4}$ or $\frac{5\pi}{4}$: P lies on the line $y = x$. That line meets the circle $x^2 + (y-1)^2 = 1$, i.e. $x^2 + y^2 = 2y$, where $2x^2 = 2x$, so at $x = 0$ or $x = 1$. The non-zero point is $z = 1 + i$.

Checking both: $z = 0$ gives $w_1 = 0 = z^*$; and $z = 1 + i$ gives

$$w_1 = \frac{1+i}{i} = 1 - i = z^*.$$

So Q_1 and Q_2 coincide exactly when P is the origin or the point $1 + i$. These are the two points where $y = x$ meets the circle, and the common positions of $Q_1 = Q_2$ are 0 and $1 - i$.

Checks

- A sample point: $P = 2i$ (the point A) gives $w_1 = \frac{2i}{2i-1} = \frac{4-2i}{5}$, and $|w_1 + i| = \left|\frac{4+3i}{5}\right| = 1$, so it lies on L.
- Numerically, we took 10^4 random points on $|z - i| = 1$ and confirmed that both $z/(z-1)$ and z^* satisfy $|w + i| = 1$ to within 10^{-9}. Scanning 2×10^5 points round the circle for $\left|z/(z-1) - z^*\right| < 10^{-3}$ found only $z = 0$ and $z = 1 + i$.

Common slips

- Dividing by z^* without first setting aside $z = 0$, and so losing the solution $P = O$.
- Concluding $z/z^* = i$ and stopping there: this fixes only the direction of P, and you still have to intersect that line with the circle.
- Getting the sign of k wrong when matching $ww^* - iw + iw^* = 0$ to the general form. The coefficient of w^* is $-k$, so $k = -i$, not i.

Beyond STEP

The map $z \mapsto z/(z-1)$ is a Möbius transformation, and such maps always send circles and lines to circles and lines. Here the image circle happens to be the mirror image of the original, which is why conjugation produces the same locus. Conjugation itself reverses orientation, so it is not Möbius, but composing it with a Möbius map gives the "anti-Möbius" maps, which include reflection in a circle. Needham's *Visual Complex Analysis* gives a beautiful geometric treatment of Möbius maps, inversion and the circle-preserving property.

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